22

GLACIERS AND GLACIATED TOPOGRAPHY

22.1 Ice and Related Phenomena

The moving ice mass downslope under the impact of gravity is called glacier. About 10 per cent of earth's surface is now covered by glaciers. About 2-3 million years BP glaciers extended over almost one-third of the earth's lithospheric surface. Glaciers are formed due to accumulation of ice above **snow line** under extreme cold climate. Snow line is generally defined as a zone between permanent and seasonal snow. In fact, snow line denotes that height above which there is permanent snow cover and thus it corresponds to the level where average temperature is always below freezing point during the warmest month of the year. The snow line is at the lowest height (0m or sea level) in the polar region and increases equatorward where it tends to occur between 5000 to 6000m. The snow line is found at the highest elevation in the drier part of the Tibetan plateau and Andes (6500m). The estimated number of all types of glaciers all over the world ranges between 70,000 and 200,000. The areas of accumulation of huge volume of ice are called **snow fields** which generate glaciers of different dimensions. The glaciers grow by gradual transformation of snow into **granular snow**, then into **firn** or **neve** and finally into **solid glacial ice.** Snow is a fluffy mass of loosely packed snowflakes of very low density having an open feather-like appearance. Semi-compacted snow due to the weight of overlying snow is transformed into granular snow of denser form. Such granular snow is called firn or neve. Further compaction of granular snow produces pure solid glacial ice.

22.2 Types of Glaciers

Glaciers are generally divided into 2 broad categories viz. (i) mountain or valley glaciers and (ii) continental glaciers. H.W. Ahlmann (1953) classified world glaciers on the basis of their thermal and morphological characteristics. He divided glaciers into 3 broad categories (continental glaciers, valley glaciers and piedmont glaciers) and into 11 subtypes as follows—

1. Glaciers without geomorphological constraints or glaciers of extensive ice sheets characterized by glacial movements in all directions

These glaciers have been further divided into 3 subtypes—

(i) Continental glaciers
(ii) Ice caps
(iii) Highland glaciers

2. Glaciers with geomorphological constraints or confined to more or less marked courses

(iv) Valey glaciers of Alpine type
(v) Cirque glaciers
(vi) Glacier tongues
(vii) Wall-sided glaciers
(viii) Transection glaciers

3. Cake-like ice sheets

(ix) Piedmont glaciers
(x) Foot glaciers
(xi) Shelf ice

Ice sheets or ice caps—The biggest glaicers on the earth's land surface are called ice sheets which are broad domes with flattened cross-section covering thousands of square kilometres. These are hundreds of kilometres in width. They submerge underlying topography. Ice radiates outward from the centre of the ice mass. These radiating ice masses are called ice sheets. The difference between ice sheets and ice caps is that of areal extent *e.g.* dome-shaped ice masses spreading over more than 50,000 km^2 are called ice sheets while small ice sheet with an area less than 50,000 km^2 is called ice cap. The central part of an ice cap or ice sheet is called ice dome. The most extensive ice sheets are Antractic and Greenland ice sheets.

Continental glaciers are in fact extensive ice sheets. These are called continental because they cover most part of the continent. Extensive ice sheets radiate outward from the centre and move downslope. During Pleistocene Ice Age extensive ice sheets moved from two centres (viz. Labrador and Keewatin) and covered about one half of North American continent. At present the biggest continental glaciers are Antarctic and Greenland ice sheets. The Antarctic continental glacier having an average thickness of 4000m covers an area of 8 million square kilometres. The ice sheets rise about 4500m above mean sea level. The Greeland continental glacier with the thickness of 3000 m at its central dome is spread over an area of 1,30,000 km^2 which is about three-fourth of the island. Smaller ice caps occur in Arctic Canada, Iceland, Norway etc.

Mountain or valley glaciers—The body of ice moving downslope under the impact of gravity through the valley bordered by rock valley walls in the mountains is called mountain glacier or valley glacier or Alpine glacier. The length of these glaciers ranges from a few kilometres to 2000 km. They are located generally above snow line as they are ablated while descending down the snowline. The important glaciers of the Himalayan regions are (figures in brackets denote the lenght of valley glaciers) : Rupal glacier (16km), Punmah glacier (27km), Rimo glacier (40km), Himarche glacier, Barche glacier, Minapin glacier (all located in the Kashmir Himalaya); Hispar glacier (60km), Baifo glacier (62km), Baltoro glacier (58km), Siachen glacier (72km), Batura glacier (58km), Sasaini glacier (158km), Mohil Yaz glacier (29km), Yaz Ghil glacier (29km), Khurdopin glacier (41.6km) (all in Karakoram range); Milan glacier (19km), Kedarnath glacier (14km), Gangotri glacier (25km), Kosa glacier (11km) (all in Kumaun Himalaya); Zemu glacier (25km) and Kanchenjunga glacier (16km) (all in Sikkim).

Piedmont glaciers—The glaciers formed due to coalescence of several mountain or valley glaciers at the foothill zone are called piedmont glaciers. Such glaciers are found only in colder areas and not in the tropical or temperate regions because they melt when they reach the foothill zone. Melaspina glacier of Alska (USA) is typcial example of piedmont glaciers.

Ice shelf is a floating thick ice sheet or ice cap attached to the coast. Since there is no friction of ice with the bed and hence ice can spread freely. Such glaciers are abundantly found along the Antarctic coasts *e.g.* Ross Ice Shelf, Ronne-Filchner Ice Shelf etc.

Ice field is comparatively flat and extensive mass of ice sheets.

Cirque glaciers—The ice occupying an arm-chair-shaped cirque in the mountains is called cirque glacier.

Niche glacier represents a small upland ice mass which rests upon a sloping rock face.

22.3 Movement of Glaciers

Glacier moves under the impact of the force of gravity which is the result of the thickness of ice mass and the gradient of the bed. In technical terms glacier advances because 'it deforms to stress set up in the ice mass by the force of gravity'. The stress is set up because of the weight of overlying ice. Hydrostatic pressure and shear stresses are the two main components of internal stress. The hydrostatic pressure representing the weight of overlying ice is the same in all directions of the ice mass. Thus, it is the shear stress which, being the outcome of the weight of the superincumbent ice and surface slope of the glacier, causes the particles of ice to slip past one another.

There are three types of movment of glaciers e.g. (i) by sliding over bedrock, (ii) by internal deformation (known as creep) of the ice and (iii) by alternate compression and extension of the ice mass. The rate of movement of glacier depends on the thickness of glacial ice and its slope *e.g.* the greater the thickness of glacial ice, and steeper its slope, the faster the movement of the glacier and vice versa. The rate of movement of glacial ice decreases from the centre line towards the sides or edges of the glaciers becasue the friction of the ice against rockwalls retards the flow from the surface towards the bottom or the bed. The

average velocity of most of the glaciers ranges between 3m and 300m per year. The internal deformation of the ice makes the glaciers sluggish in their movement. For example, Meserve Glacier in Victoria Land of Antarctica moves at the rate of 3-4m per year. On the other hand, more active basal sliding of the ice causes higher velocity of the galciers. For example, Franz Josef Glacier of New Zealand moves at the rate of 300m per year. Jakobshavn Isbrae Glacier of Greeland moves at much faster velocity (viz. 7-12 km per year). Sudden increase in the velocity of the glaciers causing velocities of 4-7km per year is called **periodic glacial surge.** The occasional glacial surge is characterized by the movement of ice at the rate of 5m per hour. The forward movement of glacier is called **advancement of glaciers** while withdrawal of glacier is known as **retreat of glacier.** The destruction of glacial front due to melting of ice consequent upon increase of temperature is called **ablation of glacier.**

22.4 Erosional Work of Glaciers

Normally, a glacier is supposed to erode the rocks, transport the eroded materials and deposit the eroded materials at suitable places like other agents of erosion and depostion but there are two contrasting view points as regards glacial erosion. One group of glaciologists, very often called as protectionists, believes that glacier instead of eroding the rocks protects them as it covers the bedrocks. This protection concept is based on the belief that glacial ice is softer than the rocks and hence it cannot erode them. The second group of glaciologists does not accept the glaciers as impotent agents of erosion rather take them as active potent agents of erosion and deposition. On an avergae, glaciers modify the pre-existing landscapes through their erosional and depositional works. The term **glacierization** means the coverage of an area by moving glacial ice while **glaciation** refers to the action of moving ice on land. It may be pointed out that glaciation may include both the processes of occupation of land by ice masses and erosional and depositional works of advancing glaciers.

The erosional work of the glaciers is accomplished through the mechanisms of **abrasion, plucking** and **polishing.** Pure ice mass is geomorphologically inactive but when coarse debris is carried by the glacier at its base it becomes active agent of erosion. Thus, the glacier erodes its bed and side walls with the help of tools of erosion (coarse debris) through the mechanism of abrasion. Large particles of well jointed rocks are detached by the moving glacial ice. This mechanism is called plucking.

22.5 Erosional and Residual Landforms

The landforms carved out of glacial erosion iclude bumps (whaleback, rock drumlin) and depressions, U-shaped valleys, hanging valleys, cirques, aretes, horns, nunataks, crag and tail, glacial stairway, Roches moutonnees, trough lakes, tarn, fiords etc.

U-shaped valleys—The cross-section of glacial valleys or glacial troughs of mountain glaciers is U-shaped which is characterized by steep valley walls with concave slope and broad and flat valley floor (figs. 22.1 and 2). Some times, U-shaped valleys are associated with tributary valleys called as **hanging valleys.** According to one group of glaciologists glaciers do not form their own valleys but flow through pre-existing fluvially originated valleys. They modify pre-existing river valleys through the mechanism of abrasion and plucking and thus they transform them into U-shaped valleys. On the other hand, the advocates of extreme glacial erosion believe that glaciers dig out their own valleys through which they flow. It may be pointed out that if we look at the depth and width of glacial valleys it becomes clear that such enormous valleys cannot be dug out by glaciers alone. It may be concluded that glacial valleys are modified forms of pre-existing valleys.

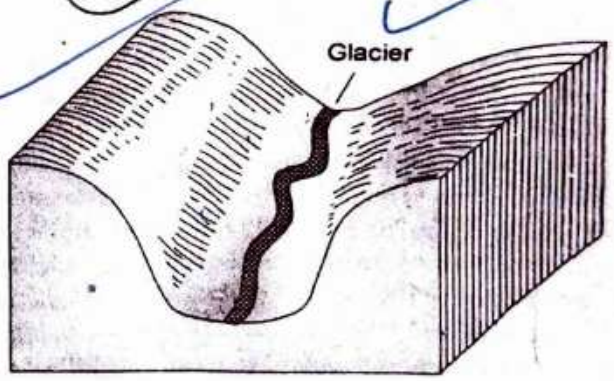


Fig. 22.1 : U-shaped valley.

Hanging valleys—The valleys of tributary glaciers which join the main glacial valleys of much greater depth are called hanging valleys (fig. 22.2) After deglaciation the meltwater of hanging valleys makes waterfalls while joining the main valley. There are two contrasting opinions about the origin and development of hanging valleys. (1) According to

glacial erosion school hanging valleys are carved out entirely by glacial erosion. The hanging valleys, according to this concept, are the result of difference between the levels of main and tributary glacial valleys because of differential over-deepening between the main and tributary valleys. The main glacier being longer, wider and more extensive than its tributaries erodes its valley more than the tributary glaciers and hence the valley of the main glacier becomes much deeper and hence the tributary valleys hang over the main valley. (2) According to **glacial protection school** the main valleys are water eroded and their tributaries originating from higher mountains carry ice with them. The main valleys are more and more deepened because of water action. On the other hand, the tributaries are unable to deepen their valleys like the main valleys because they are full of ice. Thus, the level of the tributary glacial valleys becomes higher than the main valleys. Consequently, the tributary valleys appear hanging over the main valleys. This school presents the following evidences in its support, (i) Not all the tributary glacial valleys uniformly hang over the main valleys. (ii) Not all the glacial valleys are associated with hanging valleys. (iii) Not all the tributary glacial valleys are hanging valleys. (iv) Hanging valleys are found only in those areas where tributary glaciers originate from higher height than the main glacier.

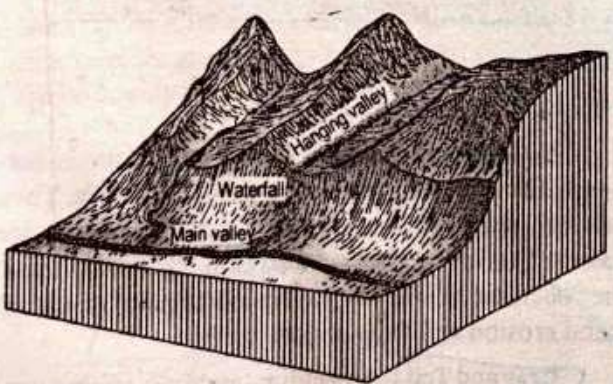


Fig. 22.2 : Hanging valley.

Cirques—The armchair-shaped or amphitheatric cirque or corrie is a horseshoe-shaped, steepwalled depression representing a glaciated valley head. Cirques are variously named in different countries e.g. **cwm** in Wales, **corrie** in Scotland, **kar** in Germany, **botn** in Norway etc. There are four main types of cirques viz. (i) simple cirques, (ii) compound cirques, (iii) hanging cirques and (iv) nivation cirques. There are 3 main components of an open cirque (fig. 22.3). (1) Head wall is almost vertical from the cirque basin and may attain the height of 600 m to 900 m with wall-like steep slope. There is no accumulation of debris at its base. It may be noted that the absence of any debris (talus) denotes the fact that there is no role of weathering in the development of cirques and thus glacial erosion is responsible for their formation. (2) The second component of cirques is their basins which represent the floors of cirques. After deglaciation the cirque basin may form lakes if water collects in it. (3) The third component is threshold which is the outer margin of cirque.

Theories of cirque formation—The following conditions favour the formation and development of cirques. (i) The water-eroded valleys should be widely spaced before the initiation of glaciation. (ii) There should be widespread intense snowfall so that snow field can be formed and glaciers may be initiated. (iii) The lithology should be homogeneous in composition. If cirques are most significant and interesting landforms of glacial erosion their origin is also complex and is still uncertain. Various geomorphologists have postulated their contrasting hypotheses to account for the origin of cirques e.g. (i) Glacial Protection Theory of Garwood, (ii) Bergschrund Nivation Theory of W.D. Johnson, (iii) Cyclic Theory of Hobbs, (iv) Meltwater and Rainwater Theory of Lewis and (v) Rotational Slip Theory.

(1) According to **glacial protection theory** of Garwood depressions are formed by frost action and water-erosion and soil creep during interglacial period before the formation of cirques (corries). These depressions are later modified by glaciers to form corries. In fact, glacial ice offers protection to corries.

(2) According to **bergschrund theory** of D.W. Johnson **basal sapping** is the most active process of corrie excavation. A crevasse or a bergschrund is caused in the glacial ice due to its movement (fig. 22.3). Alternate freeze and thaw at the interesecting point of the cirque wall and bergschrund causes mechanical disintegration of rocks and thus deepens the depression. This mechanism is called basal sapping. Gradual increase in the bergschrund and consequent acceleration of basal sapping results in the formation of well developed corrie. This theory is criticised on two main counts viz. (i) most of the bergschrunds are very small and they do not reach the rock wall or head wall and (ii) most of the glaciers contain very few bergschrunds.

(3) **The glacial erosion theory** or very often known as **cyclic theory of cirque formation** of W.H. Hobbs (1910) advocates that cirques are formed according to the various stages of the cycle of mountain glaciation. First, a small hollow is formed. Later, this hollow increases in size due to glacial erosion according to the **bergschrund theory** of W.D. Johnson. Gradually, the wall of the hollow recedes and ultimately semicircular armchair shaped depression or cirque is formed. If cirques are formed on both the sides of a mountain, they recede back and form sharp mountain peak. The intersection of receding cirques forms **cols**. Ultimately, the whole upland is dissected by the recession of corries and is transformed into fretted upland.

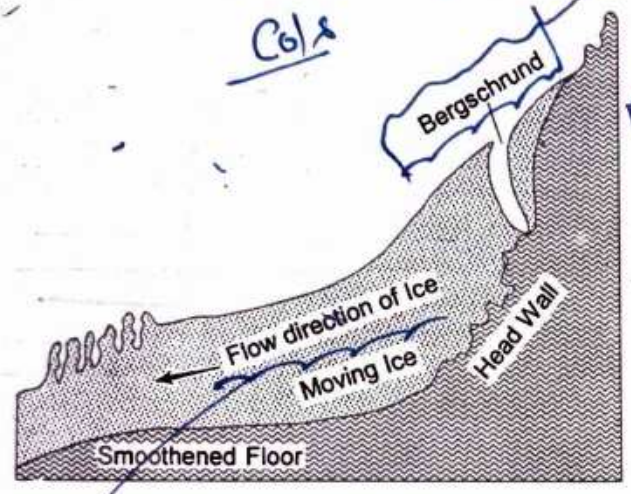


Fig. 22.3 : Cirque and its components.

Tarn—A rock basin is formed at the floor of the cirque basin due to erosion consequent upon greater thickness of ice mass and its enormous pressure. After deglaciation this rock basin is filled up with water and thus forms a small glacial lake which is called as a cirque lake or simply a tarn.

Col, aretes and horns—A high plateau or a mountain range after being eroded rather incompletely by glaciers mainly through the process of cirque recession remains 'as remnants between the steep, concave glaciated forms'. Such imperfectly glaciated upland surface is called **scalloped upland** or **biscuits board topography**. The scalloped upland is transformed into **fretted upland** by complete dissection of the terrain through cirque recession.

The mountain divide is sharpened due to recession of cirques on its both sides. Such sharpened peaks resembling saw-teeths are called **aretes**. An arete is, in fact, 'a saw-toothed divide consisting of a series of cols and intervening ridge segments.' The gap formed due to cutting of headwalls because of intersection of two steep-sided cirques is called col. Arete is called a **serrate range** in England. A pyramidal or triangular-faceted peak formed due to recession and intersection of three or more cirques is called **horn** on the basis of Matterhorn Peak of Switzerland (fig. 22.4). The crest-line of aretes and pyramidal peaks is called **col-and-peak topography**. Converging aretes looking like a star fish in plan are called **star-fish aretes**. Several examples of pyramidal peaks or horns are found in the Himalayas e.g. Mt. Godwin Austen and Broad Peak in the Karakoram Range, Badrinath and Nanda Devi in Uttaranchal Himalaya, Mt. Kailash in Tibet, Mt. Everest and Makalu in Nepal etc.

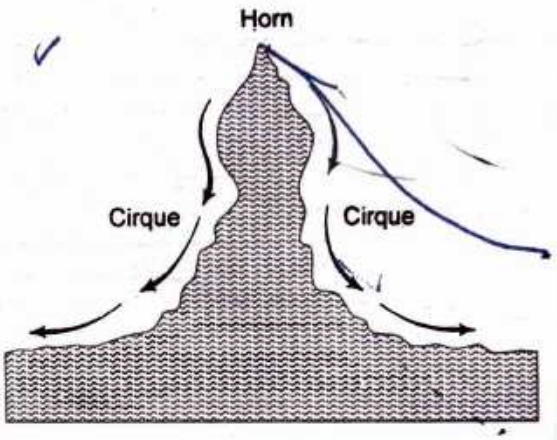


Fig. 22.4 : An example of a horn.

Nunatak—The higher peaks and mounds surrounded by ice from all sides are called nunataks. They look like scattered small islands amid extensive ice masses. That is why they are also called glacial islands. They decrease in size due to erosion caused by glacial lateral erosion and frost action.

Crag and Tail—A peculiar landform having vertical eroded steep upglacial side and tail-like appearance with lower height downglacial side is called crag and tail (fig. 22.5). Such landform is developed over old volcanic or basaltic plugs which project above the ground surface as resistant knots. These volcanic plugs offer resistance in the flow direction of glacial ice and hence the side facing the direction from which the ice comes becomes steep due to erosion and is called crag. On the other hand, the other side being sheltered by glacial ice becomes elongated with gentle slope and appears like a tail.

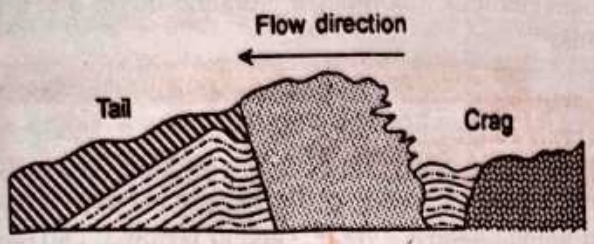


Fig. 22.5 : Crag and tail.

Roches moutonnees are streamlined asymmetrical hillocks, mounds or hills having one side smoothly moulded with gentle slope (onset or stoss side) and the steepened and craggy lee side (fig. 22.6). The term roches moutonnee was first used by de Saussure in 1804 for the streamlined rocky eminences resembling a sheep in lying posture. The onset or stoss side is smoothened through the mechanisms of abrasion and polishing by the advancing ice while the lee side is steepened due to plucking out of joint blocks by descending ice mass. There is wide range of variation in the dimension of roches moutonnees. (i) At the smallest scale they are one metre in length having ice-smoothed convex surface. (ii) At large scale roches moutonnees are several hundred metres in their dimension. They are very often aligned to structural weakness. (iii) At the largest scale roches moutonnees represent complete hills whose onset sides have been smoothed by advancing ice while the lee sides have been steepened by descending ice. Several roches moutonnees are found in the glaciated valleys of Kashmir Himalaya.

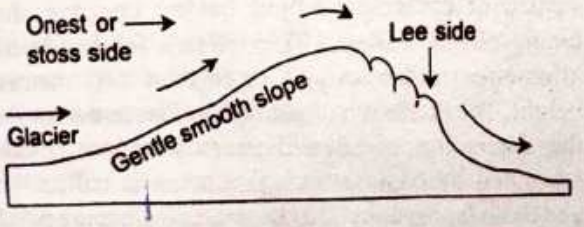


Fig. 22.6 : Roches moutonnees.

Glacial stairways also known as **giant stairways** or **cyclopean stairs**, are very picturesque and bewildering glaciated landforms. The length of each stair ranges from a few metres to several kilometres. Each stair is separated from the other by vertical cliffs measuring 30 to 300 metres. These stairs are formed in a veriety of manner. The advancing ice. of glaciers carves out giant stairway through the processes of abrasion and plucking of step faults which come across the paths of moving glaciers. The plucking of rocks at the foot of faults forms vertical cliffs. Smaller depressions are formed at the bases of cliffs. These depressions become lakes when they are filled with melt-water. These smaller lakes are called **paternoster lakes** (fig 22.7). These lakes appear as **beaded lakes** because they are associated with almost step or stair.

Glacial grooves—Small-scale streamlined depressions are called glacial grooves. Individual groove may measure 12 km in length, 100m in width and 30 m in depth.

Fiords are glacial troughs which have been occupied by the sea. In fact, fiords are the arms of the sea which have occupied U-shaped glaciated valleys which were dug out below sealevel through the mechanisms of abrasion and plucking by valley glaciers descending from coastal mountains. Fiords are characterized by steep side walls and several hanging valleys. They are very deep towards the coastal land and become shallow for some distance towards the sea but they again become deep. Thus, there is slightly raised part between the fiords and the sea. This raised part is called threshold of fiords. According to some geomorphologists these thresholds are in fact submerged terminal moraines. Most of fiords have developed during the period of intense glaciation of dissected coastal plateaus and

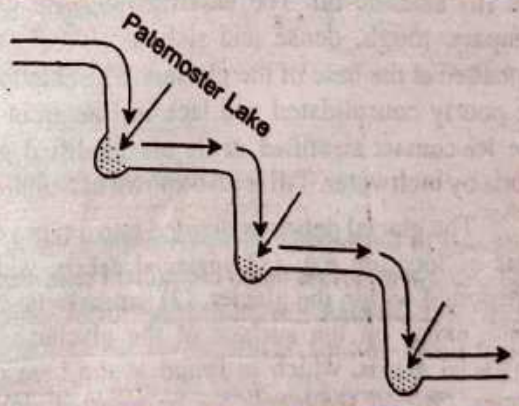


Fig. 22.7 : Glacial stairways and paternoster lakes.

mountains in Scotland. Norway, Greenland, Labrador, British Columbia, Alaska, Patagonia, New Zealand etc. The world distribution of fiords is conditioned by (A. Holmes and Doris L. Holmes, 1978) (i) tectonic structures in upland regions near the sea, (ii) pre-existing valleys which follow these structures, and (iii) heavy glaciation by seaward moving ice of sufficient thickness. There are three views regarding the origin of fiords. (1) Deep valleys were carved out by glaciers

descending from the coastal mountains above sealevel. At much later date these glaciated valleys were submerged under sea water to form fiords. (2) Fiords were formed by glacial erosion below sealevel. Sea level was significantly lowered during Pleistocene Ice Age and thus pre-existing fluvially originated valleys were further deepened by glaciers. These valleys were later submerged under sea water because of rise in sealevel due to deglaciation and these submerged glaciated valleys became fiords. (3) Fiords are supposed to have been formed due to tectonic factors. According to this concept fiords have been formed due to submergence of grabens which were formed near the coast because of faulting of the coastal land.

22.6 Transportation and Depositional Works of Glaciers

The rock debris carried by the glaciers are collectively called **glacial drifts.** Some times, the term **moraine** is used for the debris transported by the glaciers and also for the landforms made by the deposition of glacial debris. Moraine, in fact, is a collective term which should be used for depositional landforms of direct glacial origin. The glacial drifts include (1) **till,** (2) **ice-contact stratified drift,** (3) **outwash,** etc. The unsorted and nonstratified glacial drifts are called **tills** which are further divided into (i) **basal** or **lodgment till** and (ii) **ablation till.** The **basal** or **lodgment tills** are compact, tough, dense and rich in clay. They are deposited at the base of the glaciers. The ablation tills are poorly consolidated and lack in fine grain sizes. The **ice-contact stratified drifts** are modified glacial debris by meltwater. Till is also known as **boulder clay.**

The glacial debris is divided into 3 types on the basis of location e.g. (1) **englacial debris,** which is transported within the glacier, (2) **supraglacial debris,** which exists on the surface of the glacier and (3) **subglacial debris,** which is found at the base of the glacier. Glacial sediments are transported along the sides, floor and snout of the glacier. The debris falling directly into a glacier is transported without touching the bottom of the glacier. The debris falling on to the surface of a glacier is transported downslope along with the moving ice mass. The materials derived from the bed by subglacial erosion are transported by touching the bottom.

22.7 Depositional Landforms

Depositional landforms formed due to setting down of glacial drifts (glacial sediments of varying sizes) include moraines or morainic ridges and drumlins.

Moraines are ridge-like depositional features of glacial tills. They are long but narrow ridges with height more than 30m. Moraines are generally divided into 4 main categories on the basis of locational aspect of glacial deposits viz. (1) end or terminal and recessional moraines, (2) lateral moraines, (iii) medial moraines and (iv) ground moraines (see figures 8.15 and 22.8) V.K. Prest (1968) classified moraines in 3 main types on the basis of orientation of moraines with respect to ice movement :

(1) Moraines deposited transverse to ice movement

(i) ground moraines, (ii) end moraines-terminal moraines, recessional moraines and push moraines, (iii) ice-thrust moraines, (iv) ribbed moraines, (v) De Geer or cross valley moraines and (vi) linear ice-block ridge.

(2) Moraines deposited parallel to ice movement

(i) ground moraines-flutted and drumlinized. (ii) marginal and medial moraines, (iii) interlobate and kame moraines and (iv) linear ice block ridge.

(3) Non-oriented moraines

(i) Ground moraines-ablation moraines, (ii) disintegration moraines, (iii) interlobate and kame moraines and (iv) irregular ice-block ridge and rim ridge.

(1) Terminal moraines, also known as end moraines, are formed due to deposition of glacial till across the moving ice sheets at the snouts of glaciers after ablation of ice. Terminal moraines are horse-shoe shaped or crescentic ridges having concave slopes facing glacial valleys. They stretch for hundreds of kilometres in length and more than 100 metres in height. The recession of glaciers or ice sheets results in the deposition of several irregular ridges, mounds separated by basins. Such landscape is called **'knob and basin topography'**, (2) **Lateral moraines** are parallel ridges of till on either side of a glacier. They are formed due to deposition of sediments along the margins of a glacier when it contracts in size due to melting of ice. Lateral moraines are generally long, narrow and steep sided ridges parallel to the glacial valleys. They are several hundred metres in height. (3) **Medial moraines** are formed due to deposition of glacial sediments along the internal margins of two glaciers at their confluence (fig. 22.8). They project above the surface of glacial valley. (4) **Ground moraines** are formed when glacial sediments (till) are deposited at the floor

of glacial valleys. The sediments are not sorted because coarse and fine sediments are deposited together. The non-oriented moraines are usually formed due to mixture of ablation till and ice-contact stratified glacial drift.

Drumlins

The swarms of rounded hummocks resulting from the deposition of glacial till are called drumlins (fig. 22.9). They look like an inverted boat or spoon. In

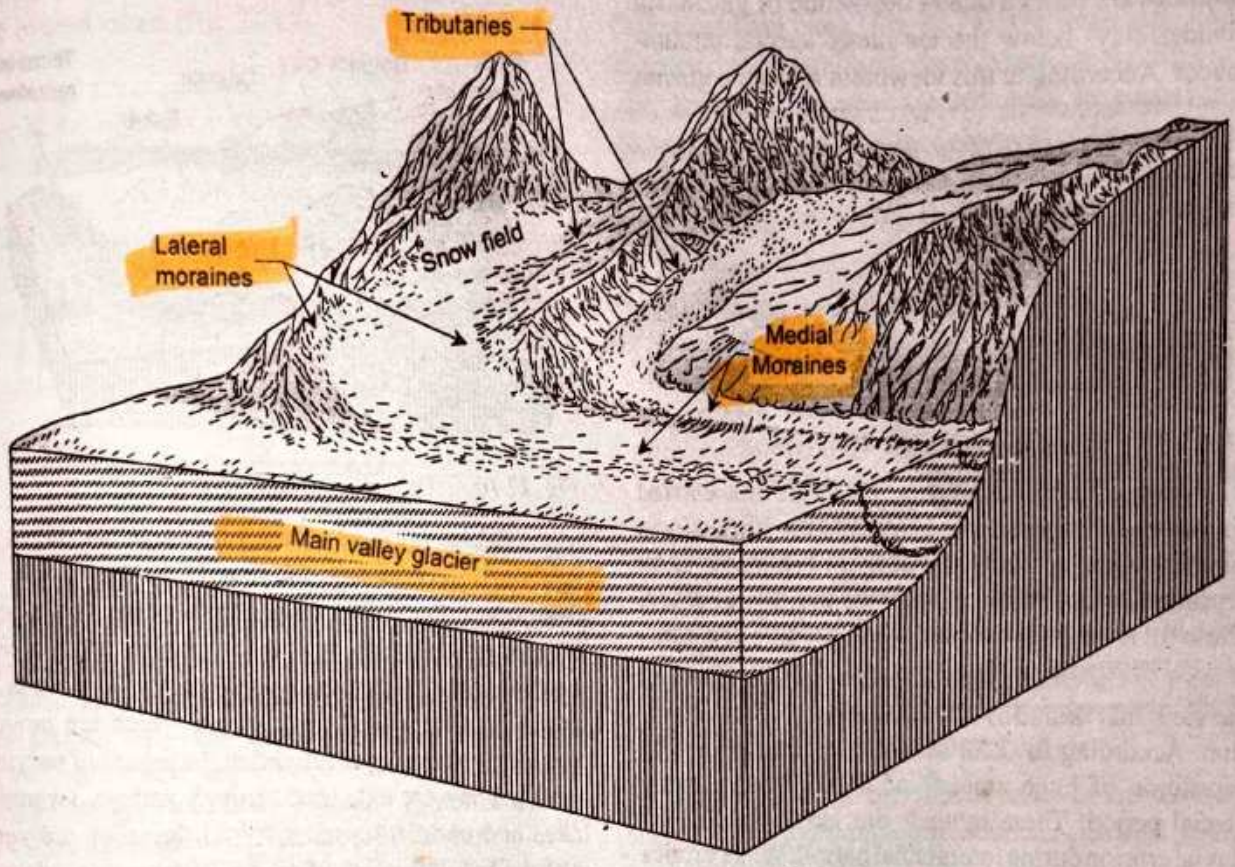


Fig. 22.8 : Different forms of morainic ridges.

fact, drumlins "are elliptical or ovoid hills, blunt on the upglacier end (steeper slope), with an elongate downglacier tail" (A. Bloom, 1978). These are streamlined hills which vary in size ranging from a few metres to 60-100 metres in height and from a few hundred metres to one-two kilometres in length. Usually, these occur in cluster and regular pattern. Such topography is called **'basket of egg topography'**. Colonies of drumlines are found in Finland, Northern Ireland and Wisconcin State of the USA where 10,000 drumlines are found in group. Drumlin clusters have also developed in middle New York and Southern Michigan states of the USA. Drainage pattern becomes irregular and indeterminant in the areas dominated by drumlin clusters. Drumlins resemble roches moutonnees in shape as both have one side with steep blunt slope and the other side with gentle slope with long tail but they differ in origin as the former are the result of deposition of glacial till (boulder clay) while the latter are erosional in character. It may be further pointed out that the upglacier side of drumline is steep while roches moutonnees' downglacier side is steep. Though most of the geomorphologists consider drumlins as the outcome of glacial deposits but some consider them as the result of fluvial erosion. The following are the prevalent viewpoints regarding the origin of drumlins.

(1) According to some geomorphologists drumlins have been formed due to modification in the terminal moraines by the glaciers. Several recessional terminal moraines parallel to each other are formed due to recession of glaciers during interglacial period. These morainic ridges are later modified by advancing glaciers during next glacial period. The advancing

glaciers erode the upglacier sides of the pre-existing terminal moraines and deposit the eroded materials on the downglacier sides. Thus, the upglacier side is steepened and the downglacier side is smoothened and lengthened (fig. 22.9).

(2) Some geomorphologists believe that drumlines are formed due to deposition of glacial till (boulder clay) below the ice under special circumstances. According to this viewpoint when the glacier

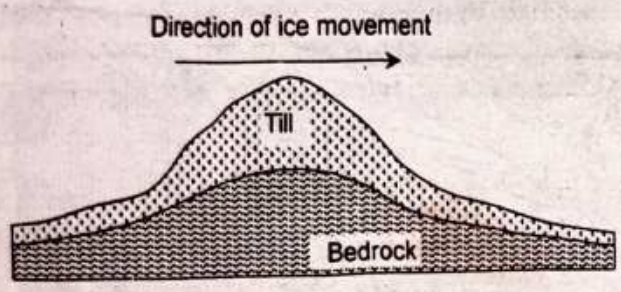


Fig. 22.9 : Drumlin.

is overloaded with sediments, these are not carried upto the snout but are deposited below the ice and form mounds. These mounds attract more sediments to be deposited because they obstruct the movement of ice. These till mounds grow in size and become drumlins.

(3) The third group of geomorphologists is of the view that drumlins are formed due to fluvial erosion. According to them mounds are formed due to deposition of huge amount of boulder clay during glacial period. These mounds are later modified by fluvial erosion during interglacial period. Water erodes the onward side of these mounds and deposit them on the leeward side. Thus, the onward side is steepened and the leeward side is smoothened and lengthened.

(4) According to Leverett drumlins are formed due to deposition of glacial till (boulder clay) by the glaciers.

22.8 Glacio-fluvial Deposits and Landforms

The snout of a glacier starts melting due to increase in temperature when it descends below **snow line.** The process of melting of a glacier is called **ablation.** Meltwater escapes through numerous but small and temporary streams. These streams carry sediments for longer distances and deposit them in various forms. These streams still carry some ice. Thus, the deposition of sediments after the ablation (melting of a glacier) is called **glaciofluvial deposit** and the landforms resulting from such deposit are called **glaciofluvial landforms.** The sediments are deposited in the form of low alluvial fans (if deposited on land) or deltas (if deposited in standing water). The fans spread out and coalesce into plains called as outwash plains. The glaciofluvial landforms include eskers, kames, kame terrace, kettle, kettle holes, outwash plains etc. (fig. 22.10).

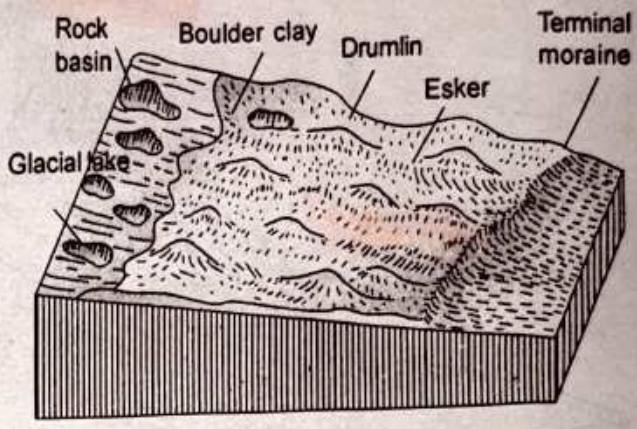


Fig. 22.10 : Depositional landforms by glaciers.

Eskers are long, narrow and sinuous ridges of sands and gravels and are situated in the middle of ground moraines. The sides of eskers are very steep. They vary in height and width ranging from a few metres to tens of metres and extend for kilometres in length parallel to the direction in which ice moved previously. Eskers are formed in the region of varying reliefs. They are extended through valleys, swamps, lakes and undulating terrains and thus they are very useful from the point of transportation as roads are easily constructed along eskers. Several hypotheses have been postulated for the origin of eskers.

(1) It is generally believed that water channel is issued from the snout of glacier due to melting of ice. This channel transports glacial fine sediments down the slope. The free flow of this channel is obstructed because of any obstacle. Consequently, sediments are deposited in the middle of the valley parallel to the main channel. Thus, long and narrow but low esker is formed.

(2) Water channel of meltwater is formed at the surface of the glacier and thus finer sediments are deposited at the ice surface. When the whole ice gets melted, sediments settle down at the floor of the glacial valley and thus esker is formed.

(3) Hollow tunnels are formed due to melting of ice at the bottom of melting glaciers. Thus, long but narrow water channels are formed below the surface of

ice of the glaciers. The sediments carried by these subglacial channels, when obstructed, settle down to form eskers. The eskers can be observed after the melting of all of the overlying ice.

Sometimes, a series of swellings are strung along the eskers at regular intervals. Such eskers are called **beaded eskers** (fig. 22.11).

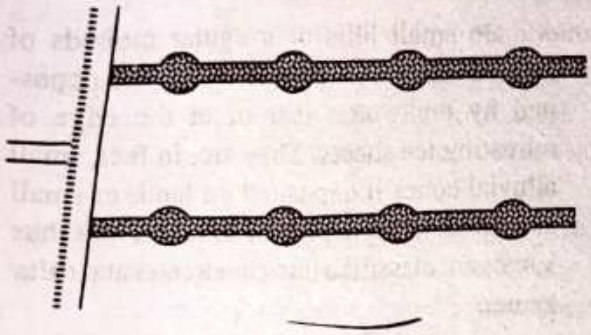


Fig. 22.11 : An example of beaded eskers.

Kames are small hills or irregular mounds of bedded sands and gravels which are deposited by meltwater near or at the edge of the retreating ice sheets. They are, in fact, small alluvial cones if deposited on the land or small alluvial deltas if deposited in the lakes. So, kames are classified into **cone kames** and **delta kames.** They are characterized by steep side slopes. Narrow flat topped terrace-like ridges formed along the trough between the glacier and the valley side are called **kame terraces.** The mounds formed in hollows and perforations in decaying ice are called **moulin kames** or **perforation kames.**

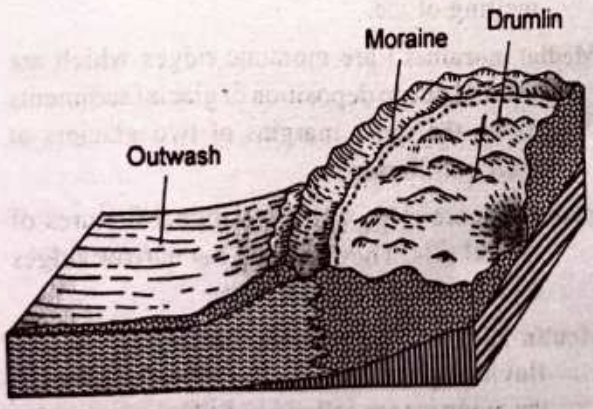


Fig. 22.12 : Moraines, drumlins and outwash.

Kettles and hummocks—Kettles are depressions in the outwash plains. Kettles are formed due to melting of large blocks of ice. Large kettles are dotted with numerous low mounds which are called **hummocks.**

Outwash—The meltwater caused due to ablation of a glacier at its snout descends through the terminal moraine and spreads like sheet water. This spreading water erodes the terminal moraines and deposits the eroded sediments in front of the terminal moraines and thus forms a plain which is called outwash plain (fig. 22.12). Outwash plains are characterized by well sorted sediments. Outwash plains also known as **'sandar'** (in Icelandic language) are characterized by multi-thread channels which are called **'braids'.** Outwash plains are, in fact, formed due to coalescence of several alluvial fans in front of terminal moraines.

22.9 Important Definitions

Aretes : Sharpened peaks resembling saw-teeths in glaciated areas are called aretes which are characterized by a series of cols and intervening ridge segments. Arete is called **serrate range** in England.

Basket of egg topography : A cluster and regular pattern of numerous drumlins in an area is called basket of egg topography.

Beaded eskers : A series of swelling are strung along the easkers at regular intervals. Such eskers are called beaded eskers.

Bergschrund : A crevasse or series of crevasses at the upper surface of mountain glacier is called bergschrund.

Biscuits board topography : Imperfectly glaciated upland surface is called **scalloped upland** or biscuits board topography.

Cirque : The armchair-shaped or ampitheatric steep walled depression representing a glaciated valley head is called cirque or **corrie, cwn** in Wales, **kar** in Germany, **corrie** in Scottland and **botn** in Norway.

Cirque glacier : The ice occupying an armchair, shaped cirque in the mountains is called cirque glacier.

Col and peak topography : The glaciated topography charactrized by crest-line of aretes and pyramidal peaks separated by cols is called col and peak topography.

Continental glacier : The glacier covering vast areas of a continent with thick ice sheets is called continental glacier.

Crag and tail : A peculiar glacial landform having vertical eroded steep upglacial side and tail-like appearance with lower height downglacial side is called crag and tail.

Cyclopean stairs : also known as **glacial stairways** or **giant stairways** are very picturesque and bewildering glaciated landforms having long (ranging from a few meters to several kilometes in length) and steep cliff face messuring 30 to 300 meters of each step.

Drumlins : The swarms or clusters of rounded hummocks resulting from the deposition of glacial till are called drumlins which look like an inverted boat or a spoon.

Eskers : are long, narrow, sinuous and steep sided ridges of sands and gravels and are situated in the middle of ground moraines.

Fiords : are glacial troughs which have been occupied by the sea. Fiords are characterized by steep side walls and several hanging valleys below sea level (under seawater).

Glacial drifts : The rock deloris carried by glaciers are collectively called glacial drifts, which include till, ice-contact stratified drifts, outwash etc.

Glacial grooves : Small-scale streamlined depressions in glaciated areas are called glacial grooves, which may measure 12km. in length, 100m. in width and 30 m. in depth.

Glacio-fluvial deposits : The deposition of sediment (glacial drifts) after the ablation (melting of glacier) is called glacio-fluvial deposit and the landforms resulting from such deposits are called glacio-fluvial landforms, like eskers, kames, hummocks etc.

Ground moraines : The moraines or glacial sediments (tills) deposited at the floors of glacial valleys are called ground moraines.

Hanging valley : The glacial valleys of tributary glaciers which join the main glacial valleys of much greater depth developed at lower height are called glacial hanging valleys.

Hummocks : Low mounds developed in an extensive kettle are called hummocks.

Ice field : is comparatively flat and extensive mass of ice sheets.

Ice sheets : Extensive glaciers on earth's land surface are called ice sheets which are broad domes with flattened cross-section covering thousands of square kilometers. In fact, radiating ice masses in all directions are called ice sheets or **ice caps.**

Iceshelf : is a floating thick ice sheet or ice cap attached to the land mass.

Kames : are small hills or irregular mounds of bedded sands and gravels which are deposited by meltwater near or at the edge of retreating ice sheets. They are, in fact, small alluvial cones if deposited on land, or small alluvial deltas if deposited in lakes, and thus kames are classified into **cone kames** and **delta kames.**

Kame terraces : Narrow flat topped terrace-like ridges formed along the trough between the glacier and the glacial valley side are called kame terraces.

Knob and basin topography : The landscape having numerous irregular ridges and mounds separated by basins formed by the deposition of glacial sediments due to recession of glaciers or ice sheets is called knob and basin topography.

Lateral morainses : are parallel ridges of tills on either side of a glacier. They are formed due to deposition of sediments along the margins of glaciers when they contract in size due to melting of ice.

Medial moraines : are morainic ridges which are formed due to deposition of glacial sediments along the inner margins of two glaciers at their confluence.

Moraines : are ridge-like depositional features of glacial tills. They are long but narrow ridges with height more than 30m.

Moulin kames : The mounds formed of glacio-fluvial deposits in hollows and perforations in decaying ice are called **moulin kames** or **perforation kames.**

Mountain glacier : The boby of ice moving downslope under the impact of gravity through the valley bordered by rock valley walls in the mountains is called mountain glacier or **valley glacier.**

Niche glacier : represents a small upland ice mass which rests upon a sloping rock face.

Nunatak : The higher peakes and mounds surrounded by ice from all sides are called nunataks.

Outwash : The plains formed in front of terminal moraines due to deposition of sediments eroded from terminal moraines are called outwash or outwash plains having well sorted sediments.

Peidmont glacier : Glaciers formed due to coalescence of several mountain or valley glaciers at the foot-hill zone are called piedmont glaciers.

Rochemoutonnees : are streamlined asymmetrical hillocks, mounds or hills having one side smoothly moulded with gentle slope (onset or stooss side) and the steepened and craggy lee side.

Sandar : Outwash plains formed by glacio-fluvial deposits in Icelandic language are called sandars which are characterized by multi-thread channels which are called **'braids'**.

Snow : is a fluffy mass of loosely packed snow-flakes of very low density having an open feather-like appearance. Semi-compacted snow due to the weight of overlying snow is transformed into **granular snow** of denser form. Such granular snow is called **firn** or **neve**. Further compaction of granular snow produces pure solid **glacial ice**.

Snow fields : The areas of accumulation of huge volume of ice are called snow fields, which generate glaciers of different dimensions.

Star-fish aretes : Converging aretes looking like star fish in plan are called star-fish aretes.

Tarn : A rock basin, formed at the floor of the cirque basin due to glacial erosion, when filled up with melt-water, is called **tarn** or **tarn lake, or cirque lake.**

Terminal moraines : also known as end moraines, are morainic ridges at the snouts of glaciers and are formed due to deposition of glacial tills across the retreating ice sheets.

Tills : The unsorted and non-stratified glacial drifts (glacial sediments) are called tills which are further divided into **basal** or **lodgement tills**, and **ablation tills**.